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On algorithmic collusion and reward–punishment schemes[☆]

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ABSTRACT

A booming literature describes how artificial intelligence algorithms may autonomously learn to generate supra-competitive profits. The widespread interpretation of this phenomenon as "collusion" is based largely on the observation that one agent's unilateral price cuts are followed by several periods of low prices and profits for both agents, which is construed as the signature of a reward–punishment scheme. We observe that price hikes are also followed by aggressive price wars. Algorithms may also converge to outcomes that are worse than Nash and penalize deviations from it. While admissible in equilibrium, this behavior throws interesting light on the relationship between high algorithmic prices and the standard mechanisms behind (human) collusion.

1. Introduction

In recent years, policymakers and researchers have worried increasingly about the alleged ability of algorithms to engage autonomously in collusive strategies. The issue deserves immediate attention, as the traditional shooting angle through which collusion between human decision-makers is prosecuted in courts usually relies on intent and communication, which do not readily apply to algorithms.

Even though some recent empirical evidence (Assad et al., 2020; Brown and Mackay, 2021) suggests that algorithmic-driven decision making may lead to high prices, to the best of the authors' knowledge, formal evidence of truly autonomous collusion by artificial intelligence algorithms (AIAs) in real life is yet to be observed. For lack of such evidence, researchers have naturally turned to simulations in synthetic environments. In a seminal paper, Waltman and Kaymak (2008) show that competing Q-learning algorithms may reach supra-competitive prices. While the authors refer to this phenomenon as "collusion", economists generally agree that genuine collusion may occur only when the high prices are sustained by a reward–punishment scheme,[1] which the authors do not test. In an influential paper, Calvano et al. (2020b) correct this omission and indeed show that the high prices are concomitant with algorithmic behavior that is observationally similar to such reward–punishment schemes. In turn, the authors refer to such algorithmic behavior as "genuine" collusion (Calvano et al., 2020b, 2023). Many recent works from academics (Parker et al., 2020; Johnson et al., 2020; Hettich, 2021; Azzutti et al., 2021; Buckmann et al., 2021; Danielsson et al., 2022; Banchio and Skrzypacz, 2022) and competition authorities (Autorité de la Concurrence and Bundeskartellamt, 2019; Competition and Market Authority, 2021) build upon this interpretation. The presence of these reward–punishment schemes, if confirmed, may arguably provide an angle for regulation of algorithmic decision-making (Calvano et al., 2020a). Indeed, while previous literature may not advocate for applying current antitrust doctrines to algorithms directly, the similarities with human collusion present an intriguing perspective worth exploring further.

In this paper, we investigate the nature of apparent reward-punishment schemes. To that aim and contrary to the literature that focuses on punishments following price cuts, we exploit the possibility of price hikes that tend towards the monopoly price. These price deviations should benefit the other cartelist and may even invite reciprocity (Rotemberg and Saloner, 1990; Mouraviev and Rey, 2011;

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[1] See e.g., Ivaldi et al. (2003) for a general survey of tacit collusion, or (Harrington, 2018) for the specific case of algorithms.

[2] Since Waltman and Kaymak (2008), we know that Q-learning algorithms often converge on prices strictly between the fully competitive and fully collusive levels. This makes the study of price hikes relevant. These hikes may be interpreted as "invitations to collude", also called "pro-collusive deviations" in Abada and Lambin (2023).

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Harrington and Zhao, 2012).[2] Instead, and rather surprisingly, we observe that they are followed by the same price wars as those that follow a price cut. We need to note that the punishment of both price cuts and price hikes is one of the many possible equilibria of the repeated game (Abreu, 1988). However, it is a rather implausible one in the context of (human) collusion, as upward price deviations serve the interests of the other agents and it would be sufficient to punish price cuts. This calls for attention regarding how algorithmic behavior may diverge from known human conduct.

The paper is organized as follows: in Section 2 we describe the apparent punishment of price hikes. Section 3 explores how algorithms may converge on outcomes worse than Nash equilibrium and appear to penalize deviations. We conclude in Section 4.

2. Experimental protocol and main findings

For ease of comparison, our study faithfully replicates the setting in Calvano et al. (2020b), which considers an infinitely repeated pricing game where two symmetric firms $i = 1, 2$ compete and adjust prices simultaneously at each time $t = 1, \ldots, T$. The agents face logit demand with an outside good. The marginal cost is set to 1 and demand parameters are such that the static Bertrand-Nash price equilibrium is $p^N \approx 1.47$ and the perfect cartel or monopoly price is $p^M \approx 1.92$. Agents delegate their price decisions to independent algorithms that are based on standard Q-learning algorithms (Watkins, 1989). This class of algorithms is often used to introduce reinforcement learning, owing to their relative simplicity and intuitiveness.[3] The algorithms have a one-period memory of past prices and use standard hyperparameters.[4] Convergence is said to be achieved and the session is stopped if the optimal strategy does not change for 100,000 consecutive periods for each agent. In any case, each session is stopped after one billion iterations. The only difference with Calvano et al. (2020b) is that we run 10,000 sessions instead of 1000. While this adjustment does not fundamentally alter any of the main outcomes, it enables us to attain greater precision when examining post-convergence deviations. We refer the reader to Calvano et al. (2020b) for further implementation details.

As in Calvano et al. (2020b), we observe that AIAs learn to set prices at an average $\tilde{p} = 1.79$, which is significantly higher than the Bertrand-Nash price equilibrium of 1.47. Rather intriguingly, we note that this price is also significantly lower than the joint-profit maximizing (monopoly) price of 1.92. It is thus natural to analyze the responses not only to price cuts but also to price hikes. To analyze responses to price cuts and price hikes, we conduct the same exercise as Calvano et al. (2020b). For each session, we start at period $\tau = 0$ and retrieve the prices on which the two algorithms have converged. While $\tau < 10$, we let the algorithms play according to their learnt strategy. Then, in period $\tau = 10$, we exogenously force one of the algorithms to lower or raise its price (by an arbitrary amount).[5] In subsequent periods, we observe the responses of both algorithms. We then average these responses over our 10,000 sessions.

We first analyze the scenario where one of the algorithms undercuts its rival, which is shown in Fig. 1(a). In line with Calvano et al. (2020b) and other papers, we confirm that price cuts are followed by several periods with low prices. This could indeed be interpreted as the signature of a reward–punishment scheme, which coheres with the hypothesis that AIAs devise such schemes to sustain supra-competitive prices and profits.

The responses to positive price deviations (price hikes, Fig. 1(b)) are, however, rather surprising: price hikes, which are unequivocally beneficial to the nondeviating agent in the short term, are also followed by aggressive price wars.[6] This pattern may be counterintuitive, and highlights a specificity of the mechanisms behind algorithmic supra-competitive prices, relative to known human conduct.

Table 1 is the counterpart of Table 2 in Calvano et al. (2020b), where we also test for price hikes. Such deviations are not tested in the original paper (the upper right part of the table is left empty). The table shows that, whatever the magnitude of the price hike, these are also systematically followed by aggressive price wars.[7]

That algorithms converging to supracompetitive outcomes may appear to punish any (upward or downward) price deviation, has also been put in evidence in Abada and Lambin (2023). The authors, however, investigate a very different setting, which is considerably more complex than that of Calvano et al. (2020b). The focus is on the dynamic optimization of battery charging, with competition in quantities. Market power does not materialize through high price, but through large price variations over time. These differences preclude a clean comparison with the literature and does not allow to ascertain that the phenomena is also at play in the seminal paper of Calvano et al. (2020b). The present paper resolves this discrepancy.

We acknowledge again that the punishment of price hikes can be part of the numerous possible equilibria of the repeated game from a theoretical standpoint. However, to the best of the author's knowledge, such a phenomenon has never been observed in real-life cases of collusion. Further, the punishment of price hikes could be viewed as the enforcement of a price ceiling which is often used as a remedy to, rather than a source of, collusive concerns (see e.g. United States Department of Justice, 1982; European Commission, 2016; Japan Fair Trade Commission, 1997).

In Section 3, we show that when the action space is shifted down so that the Nash quantity is the upper bound (and included in the action space), we also observe a systematic punishment of all deviations, including a punishment of upward deviations towards the Nash price.

3. Convergence below Nash

In this section, we replicate the simulations conducted in the main text, with one key difference: the action space has been shifted downwards. Specifically, instead of 15 actions evenly distributed between 1.42 and 1.97, we now have 15 actions evenly distributed between 1.25 and 1.47. The upper bound of 1.47 corresponds to the Bertrand-Nash price. Intuitively, it may be expected that performing algorithms would converge to the upper bound of the action space, which is the unique equilibrium of the stage game and leads to the highest feasible profit in the repeated game.

However, our results show that only 53% of the simulations converge to the static Nash equilibrium price, with 9% of the simulations converging to cycles, and the remaining 38% converging to singleton actions that are strictly below the static Nash equilibrium. Fig. 2 focuses on these simulations and presents the response to the similar stimuli as shown in Fig. 1. Consistent with our previous findings, we observe that even when the algorithms have converged below Nash, upward price deviations lead to price wars, reflecting the abnormally competitive nature of the market.

[3] While a detailed understanding of Q-learning is not necessary for the present paper, interested readers can refer to Watkins (1989) or Calvano et al. (2020b) for comprehensive descriptions of these algorithms.

[4] The learning rate is set at $\alpha = 0.15$. The exploration rate follows an ϵ-greedy rule: $\epsilon = exp^{-\beta t}$ with $\beta = 4 \times 10^{-6}$.

[5] Calvano et al. (2020b) apply the stimulus in Period 1. We have chosen $\tau = 10$ for visual clarity and evidence that the algorithms have indeed reached convergence prior to the stimulus. In a series of experiments not reported in the present paper, the stimulus has been applied to periods 0,1,10 and 20, without altering any of the results.

[6] To ensure we consider only relevant deviations, we restrict our exercises to the simulations that converge on a pre-shock price lower than the monopoly price. Similarly, we exclude price cuts when pre-shock price is already at the minimum level. In addition, we show in the Online Appendix A that responses to the stimulus do not depend on the size of the recurrent class, when algorithms converge on cycles.

[7] To ease the reading of the tables, Online Appendix B offers graphical representations of the responses to deviations in absolute prices.

(a) Replication of Calvano et al. [2020b] (Figure 4): unilateral price cut in period $\tau = 10$.

(b) Counterpart of Calvano et al. [2020b] (Figure 4): unilateral price increase in period $\tau = 10$.

Fig. 1. Both price cuts (left) and price hikes (right) are followed by price wars — average prices over all simulations.

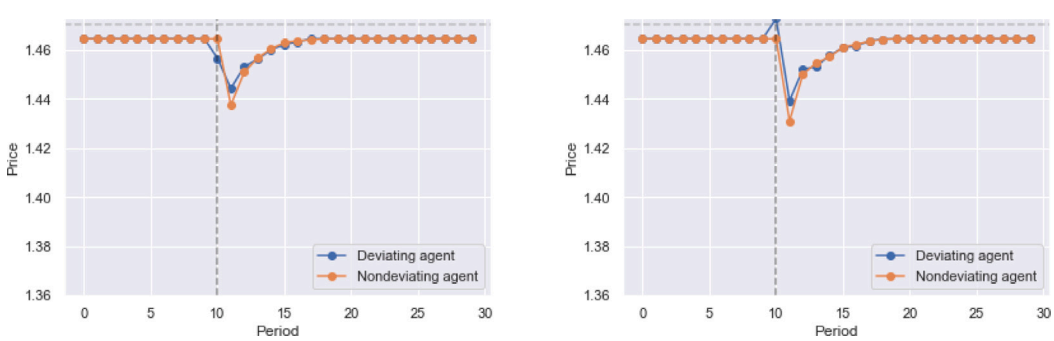
Table 1
Replication of Calvano et al. (2020b) (Table 2). An agent deviates in period $\tau = 10$. The upper-right portions of the tables (bold figures) were not shown in the original paper.

(a) Relative price change undertaken by the nondeviating agent in period $\tau = 11$ relative to $\tau = 10$.

Pre-shock	*Freq.*	*Deviation price*														
price		1.43	1.47	1.51	1.54	1.58	1.62	1.66	1.70	1.74	1.78	1.82	1.85	1.89	1.93	1.97
1.62	0.02	−0.10	−0.09	−0.10	−0.09	−0.09	0	**−0.09**	**−0.09**	**−0.08**	**−0.07**	**−0.06**	**−0.07**	**−0.05**	**−0.06**	**−0.05**
1.66	0.05	−0.10	−0.10	−0.10	−0.09	−0.09	−0.09	0	**−0.09**	**−0.08**	**−0.08**	**−0.07**	**−0.06**	**−0.07**	**−0.07**	**−0.07**
1.70	0.10	−0.11	−0.11	−0.11	−0.11	−0.11	−0.10	−0.10	0	**−0.10**	**−0.09**	**−0.08**	**−0.08**	**−0.07**	**−0.07**	**−0.07**
1.74	0.15	−0.12	−0.12	−0.12	−0.12	−0.12	−0.12	−0.11	−0.11	0	**−0.10**	**−0.10**	**−0.09**	**−0.09**	**−0.08**	**−0.08**
1.78	0.16	−0.13	−0.13	−0.13	−0.13	−0.13	−0.13	−0.13	−0.12	−0.12	0	**−0.11**	**−0.11**	**−0.10**	**−0.10**	**−0.09**
1.82	0.16	−0.14	−0.15	−0.14	−0.14	−0.14	−0.14	−0.13	−0.13	−0.13	−0.13	0	**−0.12**	**−0.11**	**−0.11**	**−0.11**
1.85	0.14	−0.15	−0.15	−0.15	−0.15	−0.15	−0.15	−0.14	−0.14	−0.14	−0.13	−0.13	0	**−0.13**	**−0.12**	**−0.12**
1.89	0.10	−0.16	−0.17	−0.17	−0.16	−0.15	−0.15	−0.15	−0.15	−0.15	−0.15	−0.14	−0.14	0	**−0.13**	**−0.13**
1.93	0.06	−0.16	−0.16	−0.17	−0.17	−0.16	−0.16	−0.15	−0.15	−0.15	−0.14	−0.14	−0.14	−0.14	0	**−0.14**
1.97	0.03	−0.18	−0.18	−0.18	−0.17	−0.16	−0.16	−0.15	−0.16	−0.16	−0.14	−0.15	−0.15	−0.14	−0.14	0

(b) Relative price change undertaken by the deviating agent in period $\tau = 11$ relative to $\tau = 10$.

Pre-shock	*Freq.*	*Deviation price*														
price		1.43	1.47	1.51	1.54	1.58	1.62	1.66	1.70	1.74	1.78	1.82	1.85	1.89	1.93	1.97
1.62	0.02	0.08	0.05	0.02	0.01	−0.02	0	**−0.06**	**−0.08**	**−0.10**	**−0.11**	**−0.12**	**−0.14**	**−0.16**	**−0.18**	**−0.19**
1.66	0.05	0.08	0.05	0.02	0.00	−0.02	−0.04	0	**−0.08**	**−0.10**	**−0.11**	**−0.13**	**−0.14**	**−0.16**	**−0.18**	**−0.19**
1.70	0.10	0.09	0.05	0.03	0.01	−0.02	−0.04	−0.06	0	**−0.10**	**−0.12**	**−0.13**	**−0.14**	**−0.15**	**−0.17**	**−0.18**
1.74	0.15	0.10	0.06	0.03	0.01	−0.02	−0.04	−0.06	−0.08	0	**−0.11**	**−0.13**	**−0.14**	**−0.15**	**−0.18**	**−0.19**
1.78	0.16	0.09	0.06	0.03	0.01	−0.01	−0.03	−0.05	−0.07	−0.09	0	**−0.13**	**−0.14**	**−0.15**	**−0.16**	**−0.18**
1.82	0.16	0.10	0.06	0.04	0.02	−0.01	−0.03	−0.05	−0.07	−0.09	−0.10	0	**−0.13**	**−0.15**	**−0.16**	**−0.18**
1.85	0.14	0.10	0.07	0.04	0.02	−0.00	−0.02	−0.05	−0.06	−0.09	−0.10	−0.12	0	**−0.15**	**−0.16**	**−0.18**
1.89	0.10	0.10	0.07	0.04	0.02	−0.00	−0.02	−0.04	−0.06	−0.08	−0.10	−0.11	−0.13	0	**−0.16**	**−0.17**
1.93	0.06	0.10	0.06	0.04	0.02	−0.00	−0.02	−0.03	−0.05	−0.07	−0.09	−0.11	−0.13	−0.14	0	**−0.17**
1.97	0.03	0.10	0.07	0.04	0.02	−0.00	−0.02	−0.04	−0.05	−0.07	−0.08	−0.10	−0.12	−0.14	−0.16	0



(a) Unilateral price cut in period $\tau = 10$.

(b) Unilateral price increase in period $\tau = 10$.

Fig. 2. Both price cuts (left) and price hikes (right) are followed by price wars — average prices over simulations having converged to points strictly below the Nash price (3831 simulations).

The systematic punishment of any deviation from a prevailing price that is below the static Nash is a possible equilibrium of the repeated game (Abreu, 1988). Nevertheless, the phenomena seems to diverge from the price levels and conduct typically associated with (human) collusion.

4. Conclusion and policy implications

One of the most pressing legal problems with algorithms is that they cannot be ascribed as having intent. Because the mere observation of high prices is not sufficient to ascertain the presence of collusion, scholars naturally began investigating algorithms' properties. Indeed, unlike with human decision-making, the decision process may be made transparent and algorithmic decisions can be simulated ex-post and at no cost (Calvano et al., 2020a). In this direction, the observation that some AI decision processes look like reward–punishment scheme seems promising. Investigating this point further, we find in our paper that such a scheme applies not only after price cuts, but also after price hikes, even when algorithms have converged below the static Nash prices. This highlights that the mechanisms behind algorithmic high prices may differ from known human conduct in meaningful ways.

In closing, it is clear that the high prices described in the present paper, as well as in previous literature, harms consumers. Whether stemming from genuine collusion or simply a failure to compete, this observation highlights the necessity for further research to understand the phenomenon and for potential regulatory intervention.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.econlet.2024.111661.

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